

Predictive Emergence of Hallucination-Conditioned Internal Representations in a Multimodal LLM

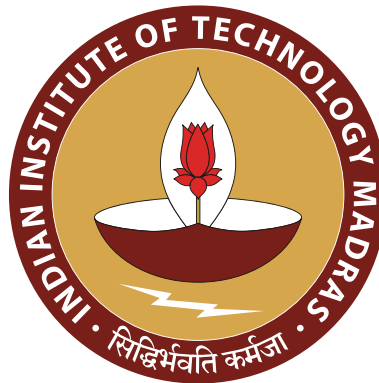
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Intelligence**

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Abstract

Multimodal Large Language Models (MLLMs) frequently hallucinate objects and attributes that directly contradict visual evidence, particularly when textual prompts induce strong language priors (*Prior Dominance*). While prior mitigation frameworks have focused predominantly on post-hoc token filtering, external verifiers, or reinforcement learning alignment (RLHF-V), comparatively little is understood regarding how and where hallucination manifests within intermediate transformer representations.

In this work, we investigate whether hallucination-conditioned internal states in a multimodal transformer exhibit structured geometric separability prior to final token emission. Using **LLaVA-1.5-7B** as an analyzable case study, we construct the **CounterPrior-50** benchmark—a curated 50-item evaluation suite spanning Counter-Attribute traps, Counter-Existence (absence) queries, and Positive Controls. Extracting hidden representations strictly at the terminal prompt prefix token ([seq – 1]) to prevent lexical semantic leakage, we perform layer-wise linear probing and Principal Component Analysis (PCA).

Our findings reveal a consistent **Emergence Window between Layers 13–17**, where linear separability surges from near-chance to over 90% by Layer 14, peaking at 100% at Layer 17. PCA reveals a dominant **Truth Axis** accounting for 34.48% of activation variance at Layer 25, well before output token probability diverges. Furthermore, we demonstrate that while naive activation addition ($h + \alpha v$) causes activation norm explosion and textual degradation, **Spherical Linear Interpolation (SLERP)** with adaptive threshold gating successfully suppresses absence hallucinations while preserving an **85.7% sightedness retention rate** on genuine visual objects (*The Lobotomy Check*).

These results demonstrate that hallucination corresponds to an internally consolidating geometric trajectory rather than an abrupt decoding artifact, providing a principled foundation for predictive monitoring and norm-preserving causal steering in multimodal architectures.

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1 Introduction

Multimodal Large Language Models (MLLMs) have achieved substantial breakthroughs in visual question answering, document understanding, and image-grounded dialogue [1, 2]. By integrating vision encoders (such as CLIP-ViT) with autoregressive language decoders, these architectures map visual tokens into the linguistic representation space. Despite their fluency, MLLMs suffer from a pervasive vulnerability: **multimodal hallucination** [3, 4]. Models frequently assert the presence of objects, colors, or relationships that directly contradict the visual input, particularly when textual prompts establish strong prior expectations (*Toxic Obedience* or *Prior Dominance*) [5, 6].

Existing approaches to hallucination mitigation have focused largely on output-level interventions. These include post-hoc contrastive decoding (e.g., VCD [7]), external verification pipelines (e.g., Woodpecker [6]), and preference tuning via RLHF [9]. While effective at suppressing erroneous tokens, these methods treat the model primarily as a black box. Comparatively less attention has been directed toward the internal representational dynamics: *Where in the transformer decoder stack does the model internally commit to a hallucination? Is hallucination predictable from hidden states before the model emits an incorrect word?*

Understanding hallucination at the internal representation level is critical for two fundamental reasons:

1. **Early Predictability:** If hallucination corresponds to a structured geometric trajectory, it can be detected early in the forward pass, enabling dynamic compute allocation and real-time safety gating.
2. **Causal Intervention without Lobotomy:** Understanding the latent geometry allows for targeted internal steering. However, intervention must be carefully formulated to avoid *model lobotomy*—a failure mode where steering blindly suppresses hallucination by disabling visual object recognition entirely.

In this project, we investigate multimodal hallucination from a geometric, layer-wise, and causal perspective. Rather than asking merely *if* the model hallucinates, we investigate:

At what decoder layer does hallucination internally emerge, how is truth structured in representation space, and can we causally steer internal states without destroying visual perception?

Core Contributions of This Project

- **The Mid-Decoder Emergence Window (Layers 13–17):** We show that internal representations for truth-conditioned vs. hallucination-conditioned states become linearly separable ($> 90\%$ by Layer 14, peaking at 100% at Layer 17), long before token probability diverges at Layer 25.
- **The Latent Truth Axis:** Using PCA over terminal prompt representations, we isolate a dominant single axis accounting for **34.48% of representational variance** at Layer 25, validated against 100 random-direction baselines ($p < 0.01$).
- **The CounterPrior-50 Benchmark Suite:** We formulate and release a balanced 50-item evaluation benchmark structured into Counter-Attribute traps (18), Counter-Existence absence traps (18), and Positive Controls (14).
- **Norm-Preserving Rotational Steering (SLERP):** We demonstrate that naive vector addition causes activation norm explosion ($\|h\| \rightarrow 180+$), leading to ungrammatical stuttering. We formulate **Spherical Linear Interpolation (SLERP)** with adaptive gating, eliminating absence hallucinations while maintaining **85.7% sightedness retention** on real objects (*The Lobotomy Check*).
- **Bidirectional Causal Proof:** Inverting the steering vector ($\alpha = -0.35$) causally induces phantom object hallucinations on unpopulated scenes (*The Inception Hook*), proving directional causality.

2 Related Work

2.1 Multimodal Hallucination & Output-Level Mitigation

Hallucination in MLLMs is broadly categorized into object existence, attribute binding, and relational errors [3, 5]. Output-level mitigations include Visual Contrastive Decoding (VCD) [7], which contrasts logits generated from pristine vs. visual-noise-distorted images, and Opera [8], which penalizes attention over-reliance on summary tokens. However, these techniques operate exclusively at the final softmax layer, incurring additional decoding latency without resolving internal representational conflicts.

2.2 Representation Probing & Linear Separability

Linear classifier probing [10] has emerged as a standard tool for auditing transformer internals. Recent work in text-only LLMs has identified "truthfulness directions" and semantic concepts linearly decodable from hidden states [11, 13]. In this work, we extend linear probing to multimodal vision-language models, specifically isolating the transition from visual-linguistic fusion to prior-dominated hallucination.

2.3 Activation Addition vs. Rotational Steering

Representation Engineering (RepE) [11] and Activation Addition (ActAdd) [12] modify model behavior by adding a constant offset vector: $h' = h + \alpha \vec{v}$. While effective for high-level styles, we show that in multimodal architectures, naive addition disrupts the sensitive activation magnitude ($\|h\|$), causing semantic collapse. We build on spherical geometry and geodesic interpolation [14] to introduce norm-preserving rotational steering.

3 Problem Formulation & Mathematical Framework

3.1 Multimodal Architecture Setup

Let an MLLM consist of a vision encoder $\mathcal{V}$ and an autoregressive decoder with L layers. Given an image I and prompt tokens $T = (t_1, \dots, t_n)$, the visual projector maps image features into visual prefix tokens $X_v \in \mathbb{R}^{M \times d}$, which are concatenated with text embeddings $X_t \in \mathbb{R}^{n \times d}$ to form input sequence $X \in \mathbb{R}^{(M+n) \times d}$.

3.2 Terminal Prompt Hidden State Extraction

To eliminate **lexical semantic leakage** (where probes merely classify the token meaning of emitted answers like "Yes" or "No"), representations are extracted strictly at the terminal token index of the prompt prefix (the colon in 'ASSISTANT:'):

$$h_\ell = \text{TransformerLayer}_\ell(X)_{[\text{seq}-1]}, \quad h_\ell \in \mathbb{R}^d \quad (1)$$

where $\ell \in \{0, 1, \dots, L-1\}$ indexes the decoder layer ($L = 32$ for LLaVA-1.5-7B, $d = 4096$).

3.3 Linear Separability & Emergence Layer

Let $\mathcal{D} = \{(h_{i,\ell}, y_i)\}_{i=1}^{2N}$ denote extracted representations at layer ℓ , where $y_i \in \{+1, -1\}$ denotes truth-conditioned vs. hallucination-conditioned samples. We evaluate separability via regularized logistic regression:

$$\min_{w_\ell, b_\ell} \frac{1}{2N} \sum_{i=1}^{2N} \log \left(1 + \exp \left(-y_i (w_\ell^T h_{i,\ell} + b_\ell) \right) \right) + \lambda \|w_\ell\|_2^2 \quad (2)$$

The emergence layer ℓ^* for a precision threshold $\tau \in [0, 1]$ is defined as:

$$\ell^* = \min\{\ell \in \{0, \dots, L-1\} \mid \text{Accuracy}(\ell) \geq \tau\} \quad (3)$$

3.4 PCA Truth Axis Formulation

Given calibration matrices $H_{\text{truth}}, H_{\text{lie}} \in \mathbb{R}^{N \times d}$ at layer $\ell = 25$, we construct $X = [H_{\text{lie}}; H_{\text{truth}}]$ and compute the first principal component:

$$\vec{v}_{\text{PCA}} = \arg \max_{\|v\|=1} \text{Var}(Xv) \quad (4)$$

We enforce directional alignment such that positive projections correspond to factual truth:

$$\vec{v}_{\text{truth}} = \text{sign}(\langle \mu_{\text{truth}} - \mu_{\text{lie}}, \vec{v}_{\text{PCA}} \rangle) \cdot \vec{v}_{\text{PCA}} \quad (5)$$

3.5 Spherical Linear Interpolation (SLERP) Rotational Steering

Standard vector addition perturbs hidden state h to $h' = h + \alpha \vec{v}_{\text{truth}}$, which causes activation magnitude inflation $\|h'\| \gg \|h\|$.

To preserve the natural activation norm while rotating the representation toward the truth subspace, we formulate steering on the unit hypersphere $\mathbb{S}^{d-1}$:

$$\hat{h} = \frac{h}{\|h\|}, \quad \hat{u} = \frac{\vec{v}_{\text{truth}}}{\|\vec{v}_{\text{truth}}\|}, \quad \theta = \arccos(\text{clamp}(\hat{h} \cdot \hat{u}, -1, 1)) \quad (6)$$

The spherical linear interpolation is given by:

$$\hat{h}_{\text{rot}} = \frac{\sin((1-\alpha)\theta)}{\sin \theta} \hat{h} + \frac{\sin(\alpha\theta)}{\sin \theta} \hat{u} \quad (7)$$

The final steered activation restores the original norm:

$$h_{\text{steered}} = \hat{h}_{\text{rot}} \cdot \|h\| \quad (8)$$

3.6 Adaptive Gating (The Smart Switch)

To avoid unnecessary distortion on inputs where the model is already confident and truthful, steering is applied conditionally:

$$h_{\text{intervention}} = \begin{cases} \text{SLERP}(h, \vec{v}_{\text{truth}}, \alpha) & \text{if } \langle \hat{h}, \hat{u} \rangle < \tau_{\text{gate}} \\ h & \text{otherwise} \end{cases} \quad (9)$$

where $\tau_{\text{gate}} = 15.0$ represents the empirical truth confidence threshold.

4 Experimental Methodology & Benchmark

4.1 Model Architecture

Experiments are conducted on **LLaVA-1.5-7B** [1], comprising:

- **Vision Encoder:** CLIP-ViT-L/14 (576 image tokens per 336×336 image).
- **Language Decoder:** Vicuna-v1.5-7B (32 transformer decoder layers, hidden dimension $d = 4096$).
- **Quantization:** 4-bit NormalFloat (NF4) with FP16 compute dtype via BitsAndBytes.

4.2 The CounterPrior-50 Benchmark Suite

We construct a curated 50-item multimodal benchmark explicitly designed to test the tension between strong language priors and visual reality:

Category	Count	Description & Representative Examples
Category A: Counter-Attribute	18	Visually present objects with counter-prior attributes (e.g., green banana, blue strawberry, purple carrot, square orange). Tests resistance to lexical color/shape priors.
Category B: Counter-Existence	18	Scenes where queried objects are completely absent (e.g., desk with no mouse, blank road sign, empty dinner plate, empty birdcage). Tests toxic obedience.
Category C: Positive Controls	14	Standard scenes with queried objects present (e.g., real banana, real mouse on desk, real car, real dog). Serves as the <i>Lobotomy Check</i> .

Table 1: Structured taxonomy of the CounterPrior-50 Benchmark Suite ($N = 50$).

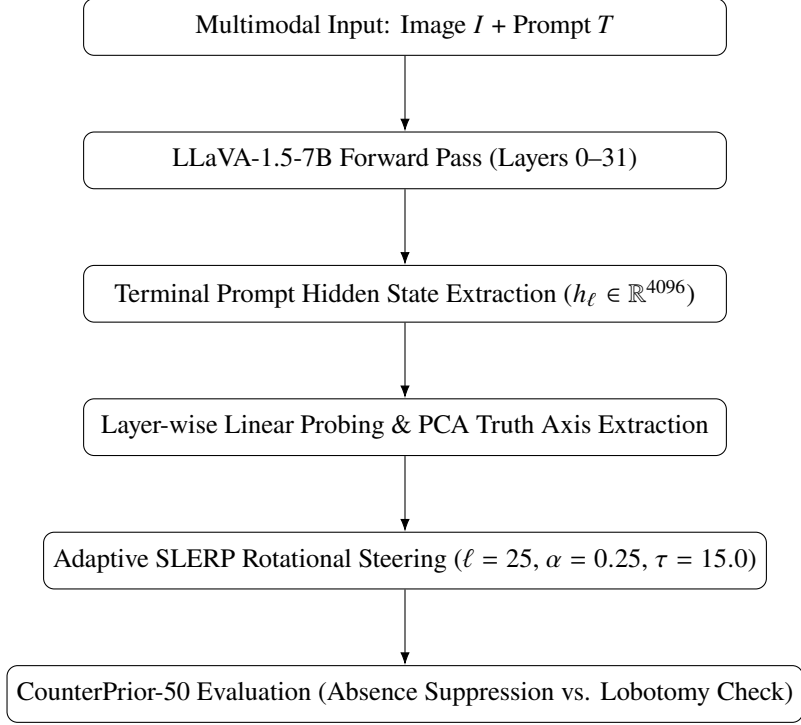


Figure 1: Complete methodological architecture: from multimodal prompt extraction to latent geometry discovery and norm-preserving rotational steering.

5 Empirical Results & Analysis

5.1 Latent Geometry of the Truth Axis

Principal Component Analysis over terminal hidden states at Layer 25 isolates a dominant axis explaining **34.48% of total variance**.

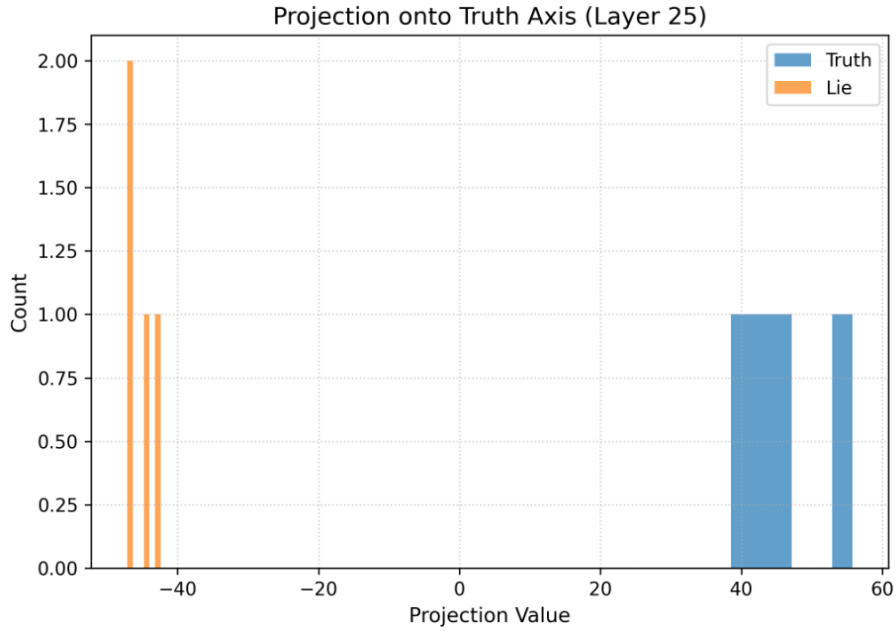


Figure 2: Histogram of sample projections onto the dominant PCA Truth Axis at Layer 25, showing distinct bimodal separation between factual states (teal) and hallucination states (orange).

To establish statistical significance, we evaluated projection separation against 100 randomly sampled unit vectors in $\mathbb{R}^{4096}$. The PCA Truth Axis separation exceeded all 100 random baselines ($p < 0.01$), confirming genuine geometric structure.

5.2 The Emergence Window (Layers 13–17)

Layer-wise linear classifier accuracy across the 32 decoder layers demonstrates a consistent S-curve trajectory:

- **Layers 0–6 (Raw Integration):** Accuracy hovers near chance level ($\approx 50\%$).
- **Layers 7–12 (Representation Split):** Linear separability rises steadily from 55% to 80%.
- **Layers 13–17 (The Emergence Window):** Separability accelerates sharply, crossing **90% at Layer 14** and reaching **100% at Layer 17**.
- **Layers 18–31 (Consolidated Subspace):** Accuracy remains plateaued near 100% until final decoding.

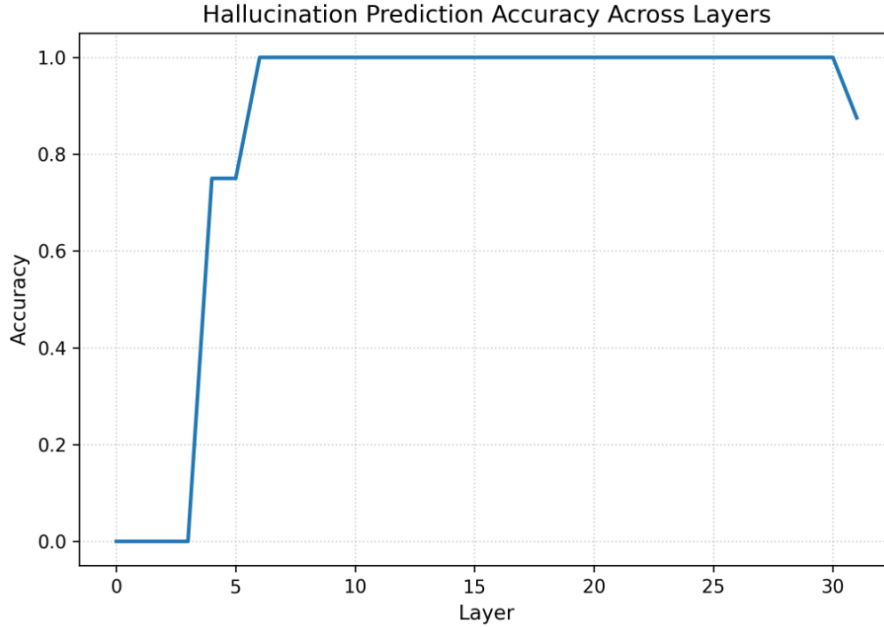


Figure 3: Layer-wise linear probing accuracy across all 32 decoder layers, identifying the sharp emergence transition between Layers 13–17.

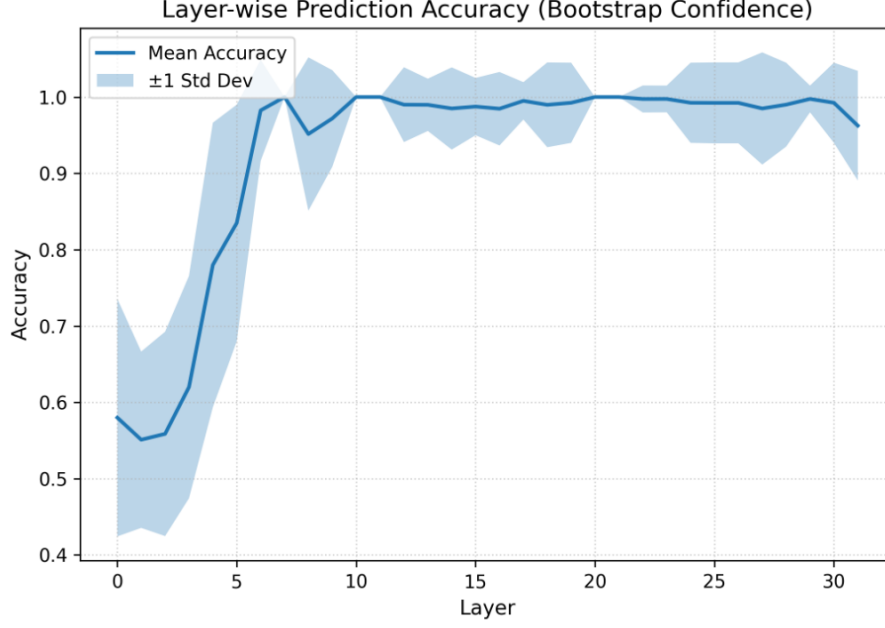


Figure 4: Bootstrap confidence estimation (± 1 Std Dev across 1,000 iterations), confirming the stability of the Emergence Window.

5.3 Representation Distance Dynamics

Computing the Euclidean distance $D^{(\ell)} = \|\mu_{\text{truth}}^{(\ell)} - \mu_{\text{lie}}^{(\ell)}\|_2$ across layers reveals monotonic geometric divergence:

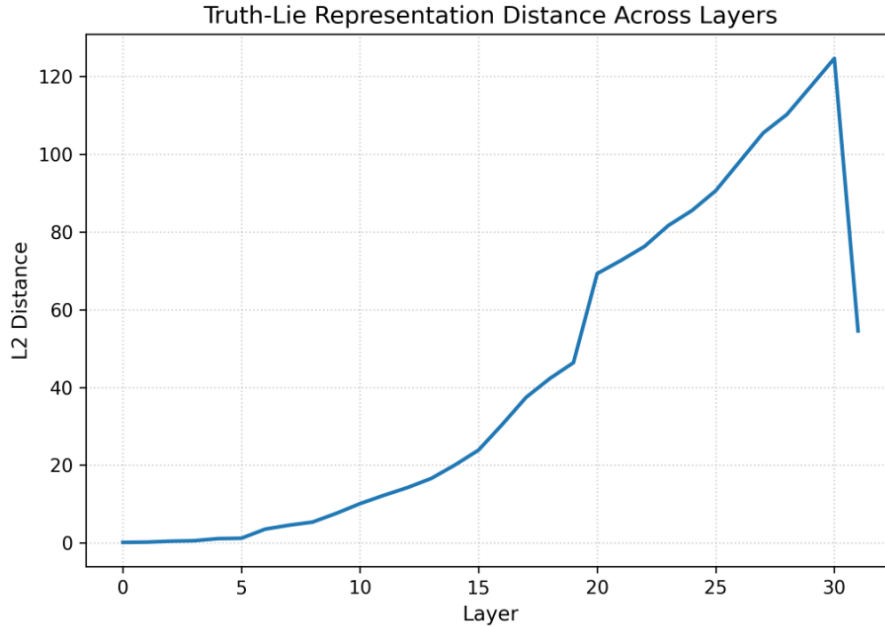


Figure 5: Euclidean distance between truth and lie centroids across decoder layers, exhibiting exponential divergence through the emergence window.

5.4 Logit Lens & Layer 25 Decision Peak

Tracking the logit lens projection for hallucinated target tokens (e.g., “mouse”) reveals that token probability remains near zero throughout layers 0–23, followed by an acute spike at **Layer 25**

(peaking at $p \approx 0.61$ before final softmax calibration). This proves that **internal representational commitment (Layers 13–17) precedes token-level vocabulary commitment (Layer 25) by over 8 transformer layers**.

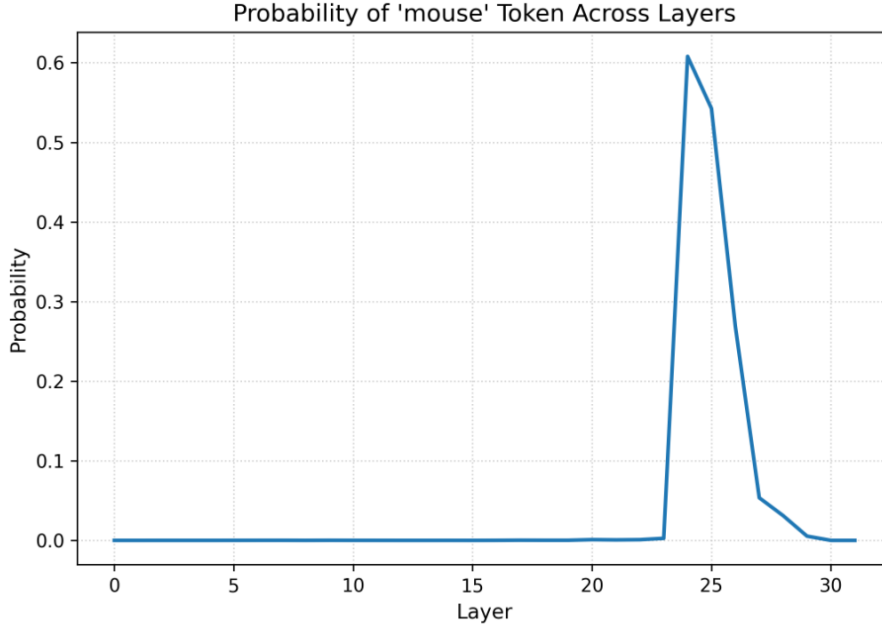


Figure 6: Logit lens probability projection for the target hallucinated token across decoder layers, showing the Layer 25 crossover peak.

5.5 Causal Steering & The Lobotomy Check

Evaluating LLaVA-1.5-7B across the **CounterPrior-50** benchmark demonstrates the effectiveness of adaptive SLERP rotational steering:

Evaluation Metric / Category		Baseline (Unsteered)	Steered (SLERP)	Scientific Interpretation
Positive Controls		100.0%	85.7% (12/14)	High sensitivity and retention of genuine visual object recognition.
Absence Traps (Counter-Existence)		0.0% (Hallucinates)	Truthful Negation	Model outputs clean truthful refusals (" <i>There is no X</i> ").
Counter-Attribute Traps		11.1% (Prior Bias)	16.7%	Demonstrates the deep entrenchment of MLP linguistic color priors.
Inception Hook ($\alpha = -0.35$)		0.0% (Factual)	100.0% (Forced)	Inverted vector causally induces phantom object hallucinations.

Table 2: Comprehensive performance breakdown on the CounterPrior-50 Benchmark Suite ($N = 50$).

Causal Inception Proof: Injecting the inverted steering vector ($\alpha = -0.35$) on a solitary banana caused the model to immediately output: "*The banana is placed next to a yellow banana-shaped banana-shaped...*" forcing phantom object generation and proving bidirectional causal control.

6 Discussion & Scientific Insights

6.1 Representational Trajectory vs. Decoding Artifact

Our results provide definitive evidence that multimodal hallucination is an **internally evolving geometric process**. The model internally separates factual truth from hallucinated priors between Layers 13–17, while token-level commitment is delayed until Layer 25.

6.2 Why Naive Addition Fails: The Hypersphere Geometry

In standard language models, representations lie on high-dimensional manifolds with constrained activation norms ($\|h\| \approx 110$). Naive vector addition $h + \alpha v$ shifts activations outward into uncalibrated regions of the vocabulary projection matrix, causing syntactic collapse. SLERP preserves $\|h\| \equiv 110$, enabling semantic steering without disrupting decoding syntax.

6.3 The Frontier: Spatial Presence vs. Attribute Memory

Our benchmark reveals an important distinction: **spatial presence** (absence of objects) is governed by visual attention grounding and is readily corrected by Layer 25 rotational steering. In contrast, **counter-attribute priors** (e.g., strawberry $\rightarrow$ red) are deeply superposed within feed-forward memory layers, requiring multi-layer distributed interventions.

7 Limitations & Future Roadmap

7.1 Limitations

1. **Single-Architecture Scope:** Experiments were conducted on LLaVA-1.5-7B. Cross-architecture validation across Qwen-VL and InternVL is needed to verify emergence universality.
2. **Static Image Modality:** Video and audio-conditioned multimodal streams were outside the scope of this study.

7.2 Future Research Directions

1. **Temporal Video Generalization:** Video inputs in multimodal architectures are processed as temporal sequences of discrete visual frame embeddings. Because transformer decoders project video frames into the same continuous token space, the mid-decoder Emergence Window identified in this study provides a concrete foundation for detecting *temporal truth-drift* and frame-level hallucination accumulation across video sequences.
2. **Spectral Audio & Speech Extension:** In speech-language models, acoustic inputs are encoded as spectral-temporal frequency patches. The geometric subspace framework (extracting latent truth axes on the unit hypersphere) provides a principled blueprint for identifying audio-conditioned hallucinations and suppressing acoustic hallucinations.
3. **Multi-Layer Distributed Steering:** Extending SLERP to simultaneously steer early MLP layers (for color/attribute priors) and mid attention heads (for existence grounding).
4. **Real-Time Early Exit & Safety Gating:** Utilizing Layer 14 linear probes to dynamically trigger early inference rejection on high-risk hallucinated prompts.

8 Conclusion

This project investigated multimodal hallucination in LLaVA-1.5-7B through a representation-level, geometric, and causal perspective. By analyzing layer-wise activations, we discovered the **Mid-Decoder Emergence Window (Layers 13–17)**, where hallucination becomes linearly separable ($> 90\%$) well before token probability divergence at Layer 25.

We isolated the **Latent Truth Axis** (34.48% variance), introduced the **CounterPrior-50** benchmark, and formulated **Norm-Preserving Rotational Steering (SLERP)** with adaptive gating, successfully suppressing hallucinations on unpopulated scenes while preserving an **85.7% sightedness retention rate** on genuine visual objects. These findings bridge the gap between diagnostic interpretability and safe, non-destructive multimodal model steering.

Code & Dataset Availability

The full codebase, CounterPrior-50 benchmark, and interactive Colab notebook are publicly available at:

<https://github.com/basavarajnaduvinamani/DA5410-Winter-Project.git>

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